习题课

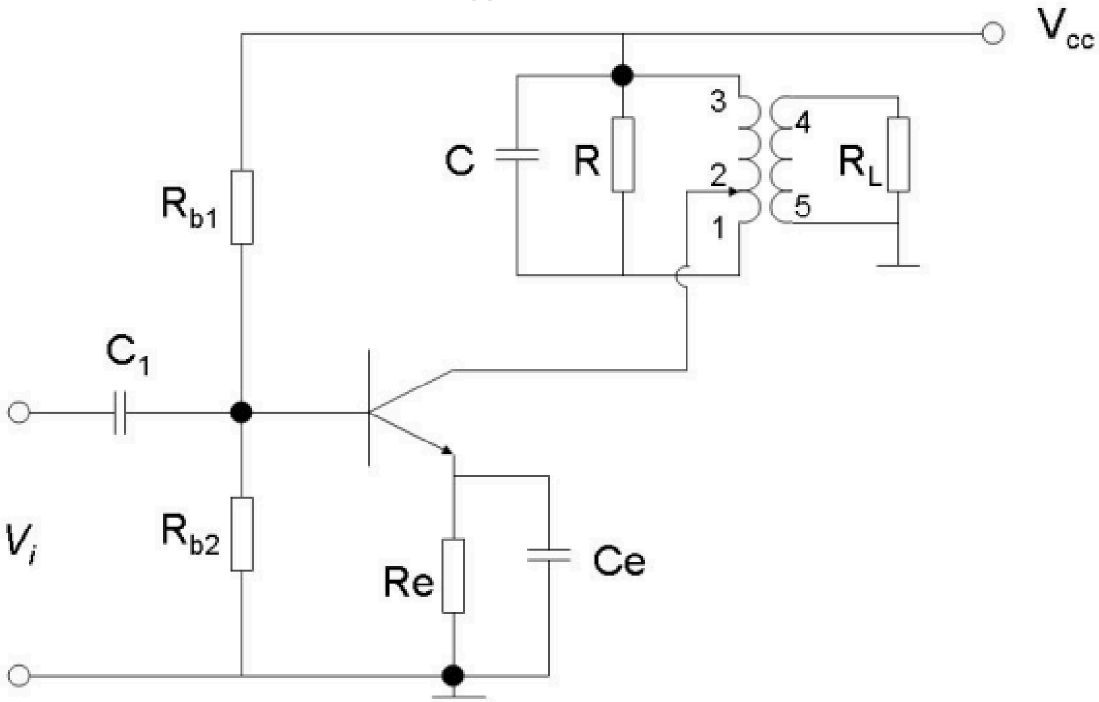
1、高频小信号放大器电路如下图所示。已知工作频率

$f_0 = 30MHz, L_{13} = 1uH$

；C与L_{13}组成的谐振回路空载品质因子$Q_0 = 100$；N_{13}=50匝，N_{23}=25匝；N_{45}=10匝；R = 10kΩ，R_L=1kΩ。三极管的y参数为：

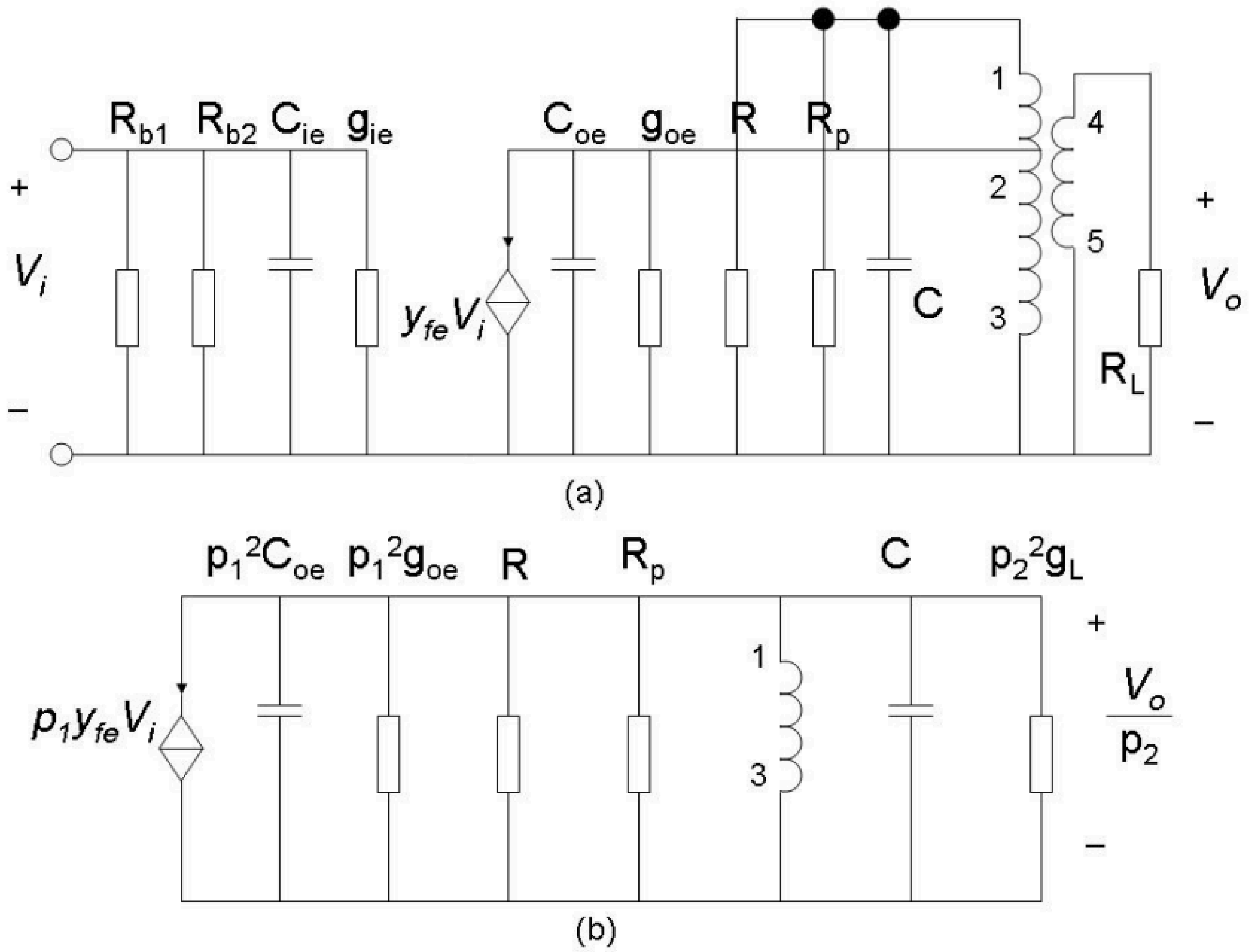
$$y_{ie} = (1.6 + j4.0)ms, y_{fe} = (36.4 - j42.4)ms, y_{oe} = (0.072 + j0.6)ms$$

忽略y_{re}



（1）画出交流等效电路（要求在图中对应标出1和3节点）；

（2）计算谐振电压增益Av_0、有载品质因子Q_L、通频带。


R_{b1}
R_{b2}
C_{ie}
g_{ie}
C_{oe}
g_{oe}
R
R_p
C
1
2
3
4
5
+
V_i
–
$y_{fe}V_i$
+
V_o
–
R_L
(a)
$p_1^2C_{oe}$
$p_1^2g_{oe}$
R
R_p
C
$p_2^2g_L$
$p_1y_{fe}V_i$
1
3
+
$\frac{V_o}{p_2}$
–
(b)

(2) $p_1 = \frac{N_{23}}{N_{13}} = \frac{25}{50} = 0.5 \qquad p_2 = \frac{N_{45}}{N_{13}} = \frac{10}{50} = 0.2$

$$G_p = \frac{1}{\omega_0 L_{13} Q_0} = \frac{1}{2\pi \times 30 \times 10^6 \times 1 \times 10^{-6} \times 100} = 53(uS)$$

$$\therefore g_\Sigma = G_p + p_1^2 g_{oe} + p_2^2 \frac{1}{R_L} = 63 \times 10^{-6} + 0.5^2 \times 0.072 \times 10^{-3} + 0.2^2 \times \frac{1}{620} = 145(uS)$$

$$\therefore |A_{v0}| = \frac{p_1 p_2 |y_{fe}|}{g_\Sigma} = \frac{0.5 \times 0.2 \times 50 \times 10^{-3}}{145 \times 10^{-6}} = 34.5$$

$$\because Q_L = \frac{1}{\omega_0 L_{13} g_\Sigma} = \frac{1}{2\pi \times 30 \times 10^6 \times 1 \times 10^{-6} \times 145 \times 10^{-6}} = 36.6$$

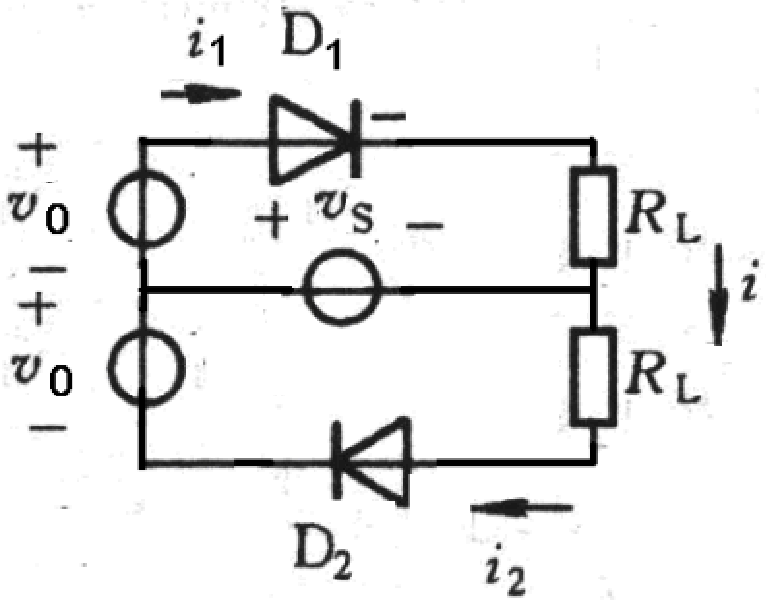
$$\therefore 2\Delta f_{0.7} = \frac{f_0}{Q_L} = \frac{30 \times 10^6}{36.6} = 820(KHz)$$

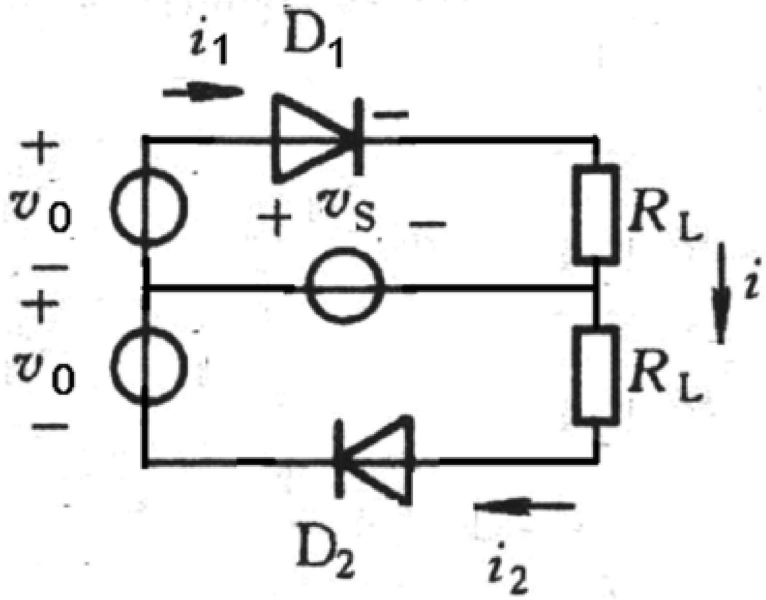
2、如下图所示，已知 $v_s = V_{sm}\cos\omega_s t$

$$v_0 = V_{0m}\cos\omega_0 t$$

$$V_{0m} >> V_{sm}$$

二极管工作在受v_0控制的开关状态，二极管**D1,D2**的伏安特性相同，均为从原点出发，斜率为g_D 的直线。推导输出电流**i**的表达式，并分析其中的频谱成分。





解： $r_d = \dfrac{1}{g_D}$

$$i_1 = \frac{1}{r_d + R_L} s(t)(v_s + v_0) \qquad \textbf{(2 分)}$$

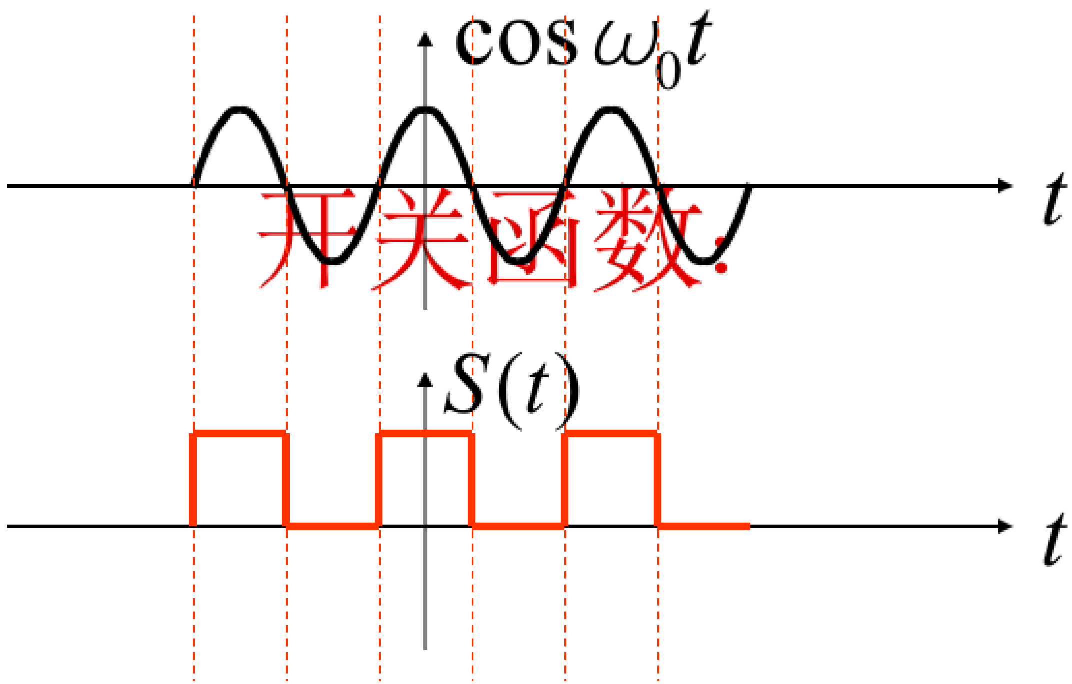
$$i_2 = \frac{1}{r_d + R_L} s(t)(v_0 - v_s) \qquad \textbf{(2 分)}$$

$$i = i_1 + i_2 = 2\frac{1}{r_d + R_L} s(t) v_0 \qquad \textbf{(2 分)}$$

$$i = 2\frac{1}{r_d + R_L}\left(\frac{1}{2} + \frac{2}{\pi}\cos\omega_0 t - \frac{2}{3\pi}\cos\omega_0 t + \cdots\right)V_{0m}\cos\omega_0 t \qquad \textbf{(2 分)}$$

输出电流的频谱成分为 ω_0**，** $2n\omega_0 (n = 1,2,3,\cdots)$ **(2 分)**

$$S(t)$$



$\because S(t)$为一周期函数,所以可以用付立叶级数展开

$$S(t)=\frac{1}{2}+\frac{2}{\pi}\cos\omega_0 t-\frac{2}{3\pi}\cos 3\omega_0 t+\frac{2}{5\pi}\cos 5\omega_0 t\cdots$$

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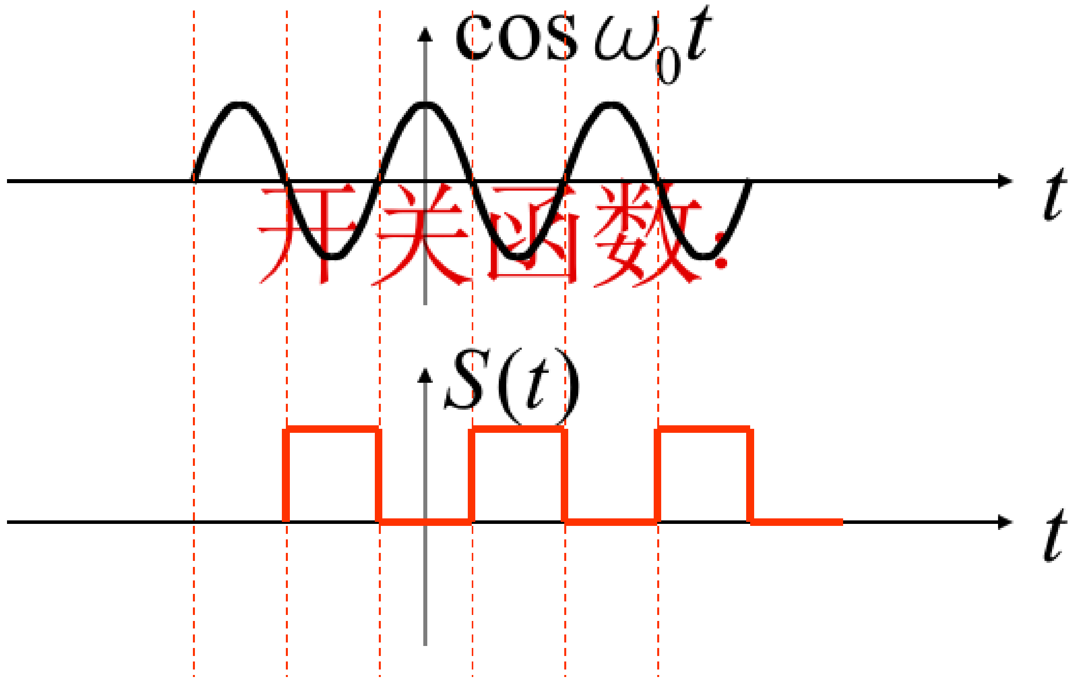
$$S^*(t) = S(t - \frac{T}{2}) = S(t + \frac{T}{2})$$



$$S^*(t) = \frac{1}{2} - \frac{2}{\pi}\cos\omega_0 t + \frac{2}{3\pi}\cos 3\omega_0 t - \frac{2}{5\pi}\cos 5\omega_0 t \cdots$$

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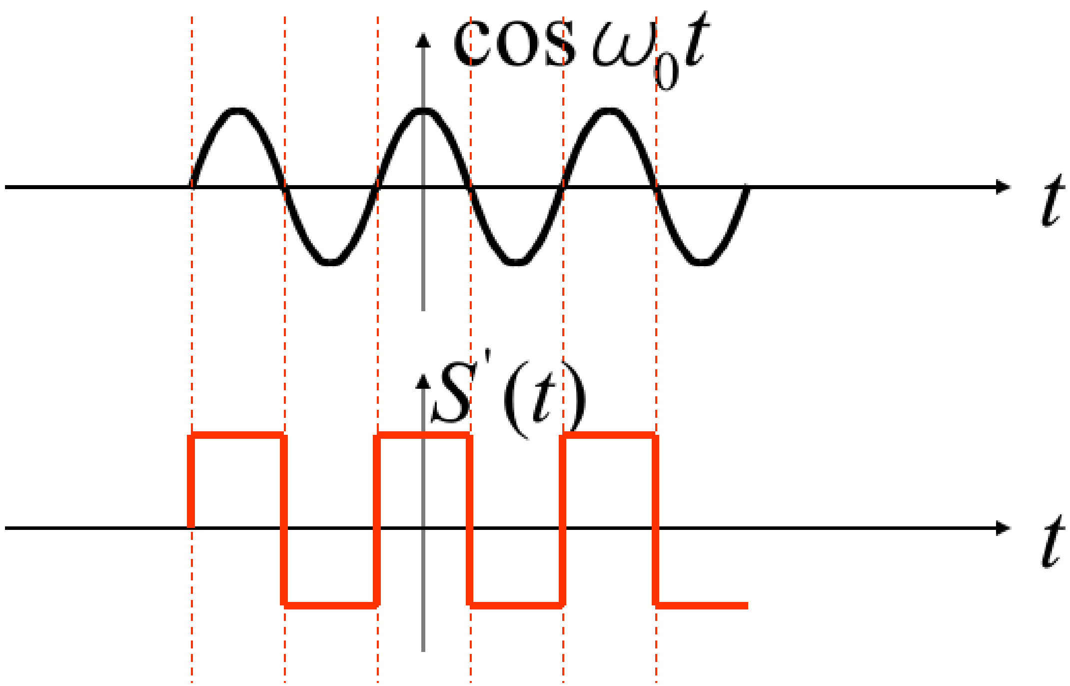
$$S'(t) = S(t) - S^*(t)$$



$$S'(t) = \frac{4}{\pi}\cos\omega_0 t - \frac{4}{3\pi}\cos 3\omega_0 t + \frac{4}{5\pi}\cos 5\omega_0 t \cdots$$

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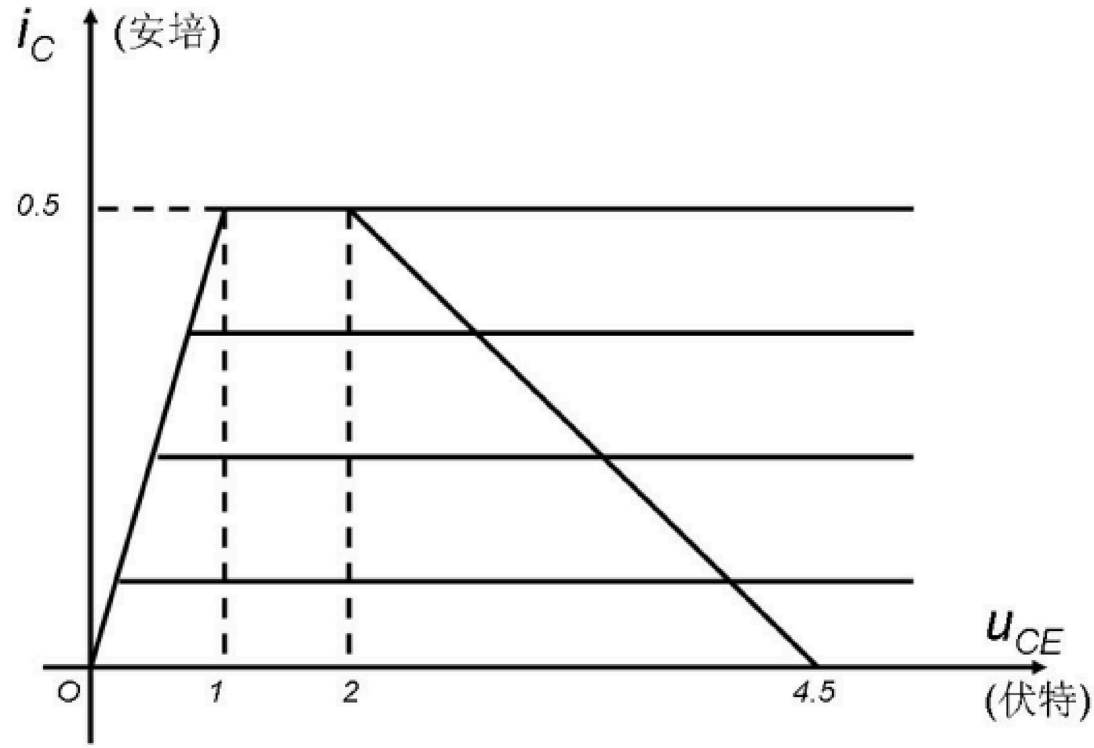
某谐振功放的动态特性曲线如图所示，并且已知$\theta_c = 90^o$，

$$\alpha_0(90^o) = 0.32 \qquad \alpha_1(90^o) = 0.5$$

（1）问该放大器工作在什么状态？求此时的**Po**、**P=**、效率、**Rp**

2）若要输出功率达到最大，问放大器应工作在什么状态？**Rp**应如何调整？调整到多大？

（调整R_p的时候θ_C不变）



(1) 放大器工作在欠压状态，从图上可读出：

$$i_{C\max} = 0.5A \quad V_{CC} = 4.5(V) \quad V_{cm} = 4.5 - 2 = 2.5(V)$$

$$I_{cm1} = i_{C\max}\alpha_1(\theta) = 0.5 \times 0.5 = 0.25(A)$$

$$I_{cm0} = i_{C\max}\alpha_0(\theta) = 0.5 \times 0.32 = 0.16(A)$$

$$P_O = \frac{1}{2} I_{cm1} V_{cm} = \frac{1}{2} \times 0.25 \times 2.5 = 0.3125(W)$$

$$P_= = I_{cm0} V_{CC} = 0.16 \times 4.5 = 0.72(W)$$

$$P_C = P_= - P_O = 0.72 - 0.3125 = 0.4075(W)$$

$$\eta_C = \frac{P_O}{P_=} = 43\% \qquad R_p = \frac{V_{cm}}{I_{cm1}} = 10\,\Omega$$

（2）若要输出功率最大，应该工作在临界状态。Rp应该增大才能从欠压变到临界。从图上看，应该使Vcm调到3.5V（此时负载线的左端恰好与拐点相交）。此时

$$R_p = \frac{V'_{cm}}{I_{cm1}} = 14\Omega$$

4、有一调幅波表达式为：

$u(t)=2\ (1+0.2\cos 2\pi\times 1000\,t+0.4\cos 4\pi\times 1000\,t)\ \cos 2\pi\times 10^6 t$伏。要求：

（1）画出调幅信号的频谱图，标出各频率分量及幅度；

（2）求频带宽度；

（3）当负载电阻为**1**欧姆时，求载波功率、各边带功率之和、以及总功率。

- (1) $\omega_0 = 2\pi \times 10^6 \quad V_0 = 2$

$$\Omega_1 = 2\pi \times 1000 \quad ma_1 = 0.2$$

$$\Omega_2 = 4\pi \times 1000 \quad ma_2 = 0.4$$

$$\frac{1}{2} ma_1 V_0 = \frac{1}{2} \times 0.2 \times 2 = 0.2$$

$$\frac{1}{2} ma_2 V_0 = \frac{1}{2} \times 0.4 \times 2 = 0.4$$

频率分量及幅度为:

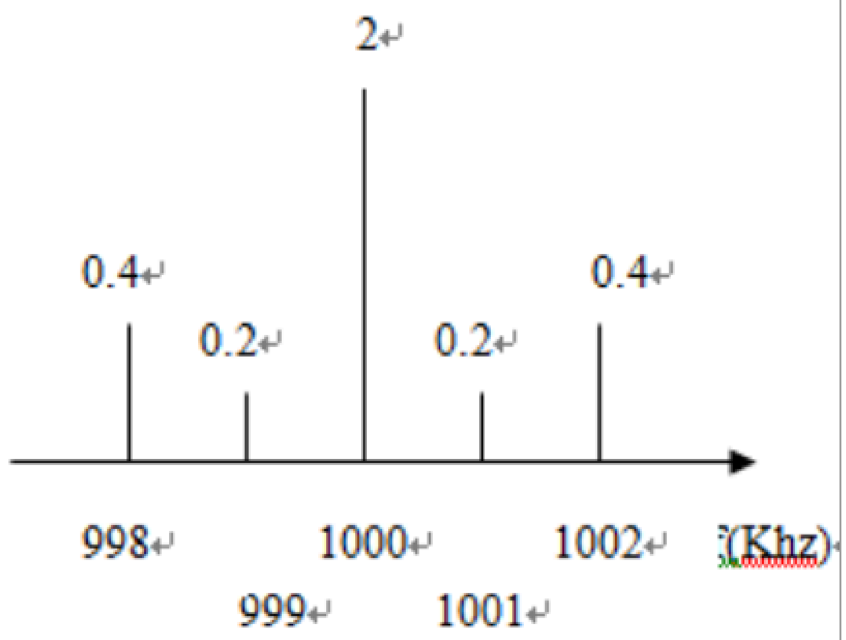
$$\omega_0 - \Omega_2 = 2\pi \times (10^6 - 2000) \qquad 0.4$$

$$\omega_0 - \Omega_1 = 2\pi \times (10^6 - 1000) \qquad 0.2$$

$$\omega_0 = 2\pi \times 10^6 \qquad 2$$

$$\omega_0 + \Omega_1 = 2\pi \times (10^6 + 1000) \qquad 0.2$$

$$\omega_0 + \Omega_2 = 2\pi \times (10^6 + 2000) \qquad 0.4$$



$$BW = 2\Omega_2 = 8\pi \times 1000$$

$$P_{OT} = \frac{V_0{}^2}{2R} = \frac{2^2}{2 \times 1} = 2(W)$$

$$P_{(\omega \pm \Omega)} = \frac{1}{2} m_{a1}{}^2 P_{OT} + \frac{1}{2} m_{a2}{}^2 P_{OT} = \frac{1}{2} \times 0.2^2 \times 2 + \frac{1}{2} \times 0.4^2 \times 2 = 0.2(W)$$

$$Po = P_{OT} + P_{(\omega \pm \Omega)} = 2 + 0.2 = 2.2(W)$$